6 The polynomial $x^3 - 4x^2 + 10x - 21$ is denoted by $f(x)$.

(a) Use the factor theorem to show that $(x-3)$ is a factor of $f(x)$. **[2]**

(b) The polynomial $f(x)$ can be written as $(x-3)(x^2+bx+c)$ where b and c are constants.

Find the values of b and c. **[2]**

(c) Show that $x = 3$ is the only real root of the equation $f(x) = 0$. **[2]**

6(a)

6(b)

$b =$

$c =$

6(c)